BENCHMARKING A HYBRID MODEL FOR MRI-BASED IDH PREDICTION AGAINST DEEP LEARNING, RADIOLOGICAL FEATURES AND AGE

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ABSTRACT

Deep-learning (DL) models predict IDH mutation from MRI but are rarely benchmarked against age and radiological features on the same patients. We compared a 3D DL ensemble (contrast-enhanced T1 [T1c] and FLAIR) with automated radiological features plus age, and with a hybrid of all three. In 1,393 adult diffuse gliomas (186 IDH-mutant; 85% grade 4; 2–21% mutant by site), the ensemble was comparable to features plus age in pooled cross-validation (AUROC 0.926 vs 0.940). It was lower on 190 reserved patients and on unseen sites (0.826–0.916 vs 0.918–0.952; UTSW: one retrained network). The hybrid had the highest AUROC in every held-out setting (0.931–0.964). In age-matched sets the ensemble retained 0.908 (age 0.41). In 359 external glioblastomas (327 confirmed IDH-wildtype), the frozen hybrid was more specific than the ensemble or features plus age (95.3%; 88.9% at the 90%-sensitivity development threshold). DL IDH models should be benchmarked against age and radiological features.

***Index Terms*—** Glioma, IDH, radiogenomics, MRI, deep learning, external validation

1. INTRODUCTION

IDH mutation status is a keystone of the integrated diagnosis of adult diffuse gliomas and requires tissue from biopsy or resection [1]. IDH-mutant gliomas generally occur in younger patients [2] and show less enhancement, higher diffusivity and lower perfusion [3, 4]. The T2-FLAIR mismatch sign is particularly associated with IDH-mutant, 1p/19q-noncodeleted lower-grade gliomas [5, 6]. DL models trained on MRI have reported AUROCs of 0.81–0.96, alone or fused with radiomics and age [7]. A T1c+FLAIR 3D ensemble trained on UCSF and UPENN reached 0.94 internally but 0.77 on external post-treatment scans [8]. Such models are seldom compared with age and radiological features on the same patients, so their added value remains unclear.

Our primary aim was to benchmark a calibrated hybrid of a 3D DL ensemble, radiological features and age against each component, under pooled cross-validation and in four held-out settings (reserved patients, two unseen sites and an external glioblastoma cohort). Secondary aims were to (i) test published imaging correlates as pre-specified hypotheses; (ii) test whether the ensemble and features plus age err on different patients, and separate the network's imaging signal from age by age matching; (iii) evaluate deferral of the least confident hybrid predictions; and (iv) examine the image information used by the network, by occlusion, attribution and probing.

2. METHODS

Data. Pre-operative MRI of adult diffuse glioma came from UCSF-PDGM [9], UTSW [10] and UPENN-GBM [11]. Two hundred patients (100 IDH-mutant) were reserved before training the models reported here, of whom 190 (92 mutant) had complete features. Excluding them, unusable masks and incomplete features, 1,393 patients remain (186 IDH-mutant; 1,185 grade 4): UCSF 433 (68 mutant), UTSW 514 (108) and UPENN 446 (10; grade 4). For external specificity we used OpenNeuro ds007045 (snapshot 2.0.1) [12]: pre-operative T1c, T2 and FLAIR with expert-validated BraTS-label masks of glioblastomas from eight centres (IDH-wildtype, WHO CNS5, per the data descriptor for its original 337 patients); 359 of 363 patients had co-registered sequences.

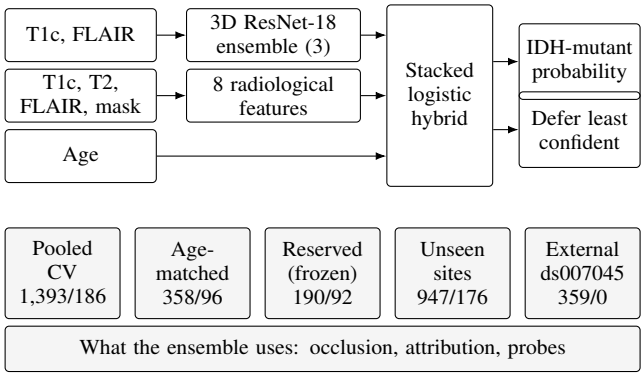


Fig. 1. Models (top) and evaluations (bottom; n / IDH-mutant).

DL ensemble. Each member pairs two 3D ResNet-18 encoders [13] reading T1c and FLAIR (128^3 tumour-centred crop), fused by cross-attention or by pooling and concatenation, with joint IDH/MGMT heads (an IDH-only head performed identically). Three members (equal weights) were trained with patient-level five-fold cross-validation (CV); each patient is scored only by the three checkpoints (one per member) that did not train on them.

Radiological features. Tumour sub-region masks came with the datasets, manually corrected (819 patients) or automated (574). Eight features were derived: log whole-lesion and enhancing volume; enhancing fraction of the lesion; minimal or absent enhancement (< 0.5 mL); necrotic/non-enhancing fraction of the core; log ratio of surrounding FLAIR abnormality to core; and two T2/FLAIR-ratio surrogates of the mismatch sign in the non-enhancing lesion (log ratio of medians; voxel fraction > 1.5), normalised to contralateral brain. Features were z-scored within site without labels. Four one-sided hypotheses (enhancing fraction, diffusivity, perfusion, T2/FLAIR ratio) were tested with site-stratified permutation tests (Holm). Diffusivity was ADC (UCSF) or DTI trace (UPENN); perfusion was ASL-CBF (UCSF) or DSC-rCBV (UPENN).

Models and evaluation. L2 logistic regressions combine age, the features and/or the ensemble logit; the *hybrid* uses all three. A parameter-free confidence gate weights features plus age and the

Table 1. AUROC by setting (95% CI for pooled CV) and specificity (%) in ds007045 at frozen thresholds (Youden / 90% development sensitivity). Pooled CV: repeated five-fold, all sites seen in training. Age-matched: AUROC within 96 matched sets (each mutant against its same-site controls within 2 years). Reserved: frozen models, 190 patients set aside before training (case-control enriched); grade 4: its 127 grade-4 patients. Unseen site: networks and combiners retrained without that site; UTSW: one retrained network. Site-seen row: the same network(s) in pooled CV, scored on that site. –: not computed. Bold: column maximum (site-seen row excluded).

Model (n / mutant)	Pooled CV (1,393 / 186)	Age-matched (358 / 96)	Reserved (190 / 92)	Reserved, grade 4 (127 / 31)	UCSF unseen (433 / 68)	UTSW unseen (514 / 108)	ds007045 spec. (%) (359 / 0)
Age	0.889 (0.864–0.913)	0.412	0.887	0.900	0.896	0.867	87.7 / 61.0
Features	0.883 (0.854–0.909)	0.878	0.870	0.756	0.904	0.847	85.5 / 55.4
Features + age	0.940 (0.923–0.956)	0.859	0.943	0.933	0.952	0.918	87.7 / 78.6
Ensemble	0.926 (0.904–0.945)	0.908	0.879	0.834	0.916	0.826	90.8 / 79.9
site seen in training					0.947	0.882	
Ensemble + features	0.933 (0.911–0.952)	**0.924**	0.915	0.856	0.934	0.882	93.6 / 87.7
Ensemble + age	0.955 (0.938–0.971)	0.908	0.943	0.943	0.961	0.913	90.8 / 88.0
Hybrid (all three)	**0.961** (0.946–0.974)	**0.924**	**0.953**	**0.952**	**0.964**	**0.931**	**95.3** / 88.9
Confidence gate	0.956 (0.940–0.971)	0.912	0.944	0.927	–	–	93.6 / **91.4**

ensemble by their distance from 0.5. Combiners are stacked on out-of-fold ensemble logits and evaluated with site- and class-stratified five-fold CV repeated five times, with Youden thresholds chosen in training folds. 95% CIs come from 2,000 stratified bootstraps (paired for differences; exact binomial for specificity); paired specificities are compared with exact McNemar tests. Calibration is summarised by the logistic recalibration slope, which case-control sampling does not affect; the hybrid's coefficients are standardised, with bootstrap CIs. *Held-out settings.* (a) Frozen models (ensemble = mean of the 15 fold checkpoints) and thresholds were applied once to the reserved patients, with features z-scored by development statistics. (b) For UCSF, all three members were retrained on UTSW and UPENN with inner five-fold CV and the combiners fitted on their inner out-of-fold logits; for UTSW, one member (cross-attention) was retrained likewise. (c) Frozen models and thresholds (Youden, and 90% sensitivity in development) were applied to ds007045, with features z-scored within the cohort (Fig. 1).

Age. In a pre-declared age-matched sample, each mutant was matched to up to three wildtype patients from the same site within 2 years; AUROC was computed within matched sets, with a set-level bootstrap. The information unique to each input is the AUROC of its cross-fitted residual after regression on the other two. The least confident hybrid predictions (log-odds distance from the in-fold threshold) are deferred [14].

What the network uses. Out-of-fold checkpoints scored each patient with occlusion of everything but the lesion (with or without a 10 mm margin), of the lesion and margin, or of T1c or FLAIR, filled with the normalised background or, as a fill control, blurred ($\sigma = 4$ voxels). Pre-declared shape controls replaced the occluded region with a ± 1 SD silhouette or inpainted it, and 11 lesion-geometry features (size, shape, position) were modelled as above. Occluded inputs are out of distribution, so only AUROC is interpreted. In 131 outcome-balanced cases (TP, TN, FP, FN), integrated-gradient attribution [15] was summarised by region. Ridge probes [16] decoded the features from the 256-dimensional vector read by the IDH head.

3. RESULTS

Imaging correlates. All four hypotheses were supported (Holm $p < 0.001$), in the same direction at every site tested (diffusivity and perfusion at UCSF and UPENN only). Within-site AUROC was 0.86 for enhancing fraction ($n = 1{,}393$), 0.79 for diffusivity (821), 0.75 for perfusion (783) and 0.64 for the T2/FLAIR ratio (1,393); within site and grade, AUROCs were 0.66–0.75, and enhancing fraction was inconsistent across sites in lower-grade strata.

Pooled CV. The ensemble and features plus age did not differ detectably (-0.015, -0.034 to 0.004; Table 1). The hybrid improved on both ($+0.035$, 0.021–0.050; $+0.020$, 0.011–0.031) but not clearly on ensemble plus age ($+0.006$, -0.001 to 0.013); it was well calibrated (slope 0.99, Brier 0.042) with 86.6% sensitivity and 92.2% specificity, and resolved 179 of 236 cases where exactly one component was right. The gate performed similarly to the hybrid (-0.005, -0.010 to 0.001). In the hybrid, the ensemble logit (odds ratio per SD 4.18, 3.16–6.27) and age (0.26, 0.19–0.32) had the largest weights, and six of the eight features had CIs excluding 1.

Held-out settings. Whenever training excluded the test patients or their site, features plus age outperformed the ensemble (Fig. 2a): reserved $+0.065$ (0.018–0.117; post hoc), UCSF $+0.035$ (0.002–0.071) and UTSW $+0.092$ (0.044–0.143). With UCSF or UTSW unseen, the ensemble lost 0.031 (members -0.001 to -0.033; the ensemble, 0.916, fell below its best member, 0.922) and the single UTSW network 0.056 relative to pooled CV, whereas, relative to the same sites in pooled CV (0.958 and 0.918; 0.974 and 0.946), features plus age lost at most 0.006 and the hybrid 0.010 and 0.015. The hybrid minus features plus age was $+0.010$ (-0.008 to 0.028) reserved, $+0.013$ (-0.002 to 0.031) UCSF and $+0.013$ (0.005–0.023) UTSW; minus ensemble plus age, $+0.010$ (-0.003 to 0.023), $+0.003$ (-0.010 to 0.017) and $+0.017$ (0.004–0.034). On the reserved set the hybrid had 88.0% sensitivity and 87.8% specificity at the frozen threshold, and a calibration slope of 1.03 (0.82–1.41; gate 1.39, 1.08–2.01). At the frozen 90%-sensitivity thresholds, reserved sensitivity was 92.4% (84.9–96.9) for the hybrid, 94.6% (87.8–98.2) for features plus age and 81.5% (72.1–88.9) for the ensemble; among grade-4 mutants, 29, 29 and 22 of 31. Reserved grade 2–3 tumours were almost all mutant (61 of 62), and the reserved set had a larger share of grade-4 mutants than development (34% vs 21%); within its 127 grade-4 patients, features plus age led the ensemble by 0.100 (0.018–0.194; Table 1).

Age matching. Within IDH class the ensemble was weakly associated with age ($\rho = -0.09$ wildtype, -0.18 mutant). Within the 96 age-matched sets, age scored 0.41 (0.34–0.49), the ensemble 0.908 (0.867–0.947), the features 0.878 and features plus age 0.859

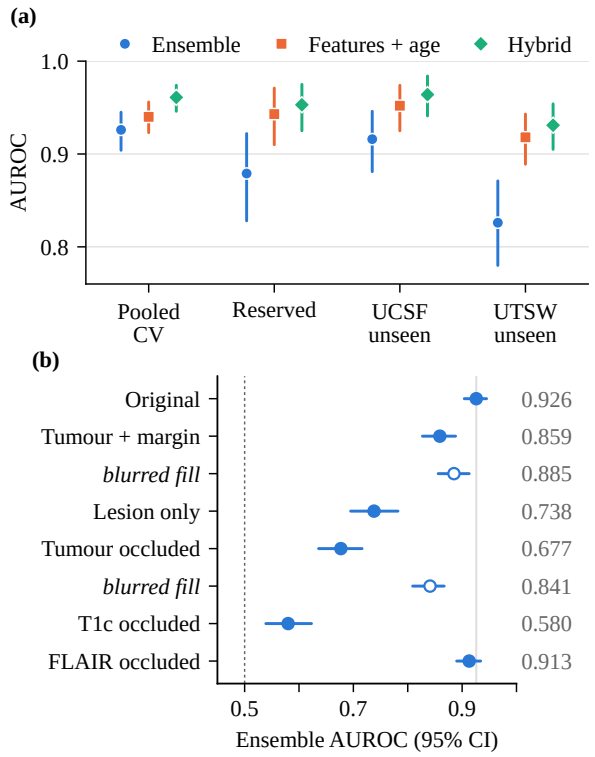


Fig. 2. (a) Ensemble, features plus age and hybrid in pooled CV and when the test patients or their site were held out of training. (b) Ensemble AUROC with parts of the input occluded (hollow: blurred instead of background fill); dashed line: chance; grey line: original.

(Table 1). The ensemble minus the features was $+0.031$ (-0.016 to 0.075), and adding the ensemble to the features gained $+0.046$ (0.012–0.082). Matching retained the older half of mutants (median age 48 vs 32 unmatched). The residual AUROC was 0.699 for the ensemble beyond age and features, 0.702 for age and 0.562 for the features.

External specificity. In ds007045, at the frozen Youden thresholds, the hybrid made 17 false-positive calls (specificity 95.3%, 92.5–97.2), the ensemble 33 (90.8%) and features plus age 44 (87.7%, as did age alone); at the 90%-sensitivity thresholds, specificity was 88.9%, 79.9% and 78.6% (Table 1). The hybrid differed from features plus age ($p < 0.001$) and from the ensemble ($p = 0.003$ and < 0.001) at both operating points (McNemar); the ensemble did not differ from features plus age ($p = 0.15$ and 0.68). Mean predicted probability was 0.037 (hybrid) and 0.065 (features plus age). With features z-scored within centre, hybrid specificity was 95.5%.

What the ensemble uses. Occluding everything but the tumour and a 10 mm margin retained AUROC 0.859 (0.885 with a blurred fill; lesion without margin 0.738); occluding the tumour and margin retained 0.677 with a background fill and 0.841 with a blurred fill (Fig. 2b). The tumour's silhouette alone scored 0.525. With the tumour occluded, inpainting gave 0.678, and the tumour-occluded output added 0.015 (0.004–0.026) beyond lesion geometry (0.792). Occluding T1c reduced AUROC to 0.580, and a hybrid refitted on the T1c-occluded logit scored 0.940, equal to features plus age; occluding FLAIR cost 0.013 (-0.025 to -0.002). A median 63% of positive T1c integrated-gradient attribution fell more than 10 mm from the lesion, close to that region's 70% volume share, although occluding that region cost only 0.04–0.07 AUROC. Within IDH class, the ensemble logit tracked enhancing fraction ($\rho \approx -0.45$), not the T2/FLAIR ratio, diffusivity, perfusion or age ($|\rho| \leq 0.2$). Enhancement was decodable from the IDH head's input (R^2 0.27–0.30); the T2/FLAIR ratios were not ($R^2 \approx 0$). Networks reading T2 and FLAIR scored 0.73–0.78; they added 0.010 (0.002–0.018) to features plus age but nothing to the hybrid ($+0.001$).

Deferral. Deferring the 20% least confident hybrid predictions captured 69% of errors and raised accuracy on the rest from 0.915 to 0.967 (sensitivity 94.4%, specificity 97.0%); error among retained cases was 0.033 for the hybrid and 0.039 for the gate (AURC difference -0.0042, -0.0081 to -0.0003).

4. DISCUSSION

The ensemble matched age and radiological features only at sites seen in training. When the test patients or their site were held out, features plus age scored higher in every setting and, unlike the networks, lost little accuracy at unseen sites. Re-selecting members inside training folds cost only 0.006, and the ensemble's pooled AUROC (0.926) lies within the range reported for DL models [7, 8], so neither member selection nor an unrepresentative network is likely to explain the gap. Grade almost separates the reserved classes, so reserved-set AUROCs largely reflect grade; the gap was nonetheless larger among the 127 reserved grade-4 patients. On these patients the hybrid stayed calibrated (slope near 1), whereas the gate was under-confident (slope above 1).

The hybrid benefits from the two components erring on different patients. In pooled CV, features plus age led in grade 4 (0.931 vs 0.878; difference 0.053, 0.004–0.108). The ensemble tended to lead under 40 (0.923 vs 0.868; -0.005 to 0.118) and in grade 2–3 (0.814 vs 0.789; -0.047 to 0.090). The hybrid was best or within 0.01 of the best in every subgroup. In age-matched sets, where age no longer separated the classes (0.41, slightly reversed), the ensemble kept its accuracy and added to the features, supporting a contribution beyond age. Externally, the hybrid was more specific than the ensemble or features plus age at both operating points. Because ds007045 has no mutants, this may partly reflect a shift towards wildtype calls, as the hybrid's lower mean probability suggests. However, at the 90%-sensitivity threshold the hybrid detected 29 of 31 grade-4 reserved mutants, as many as features plus age. Each input kept independent weight in the hybrid, although once the ensemble and age were combined the features added little discrimination.

The network's added information depended mainly on T1c. With T1c occluded, the refitted hybrid equalled features plus age. Within the tumour and margin, the network relied on image content, as the silhouette alone scored near chance. With the tumour occluded, inpainting, which removes the outline, performed like background fill. The remaining signal added little beyond lesion geometry. Integrated-gradient attribution tracked regional volume and disagreed with occlusion, so it was a poor guide to what the network used. The association with enhancing fraction and the probe results suggest that enhancement-related information contributes, without identifying which aspect. Without T2, the network could not assess the mismatch sign; a T2-reading network improved features plus age but not the hybrid.

5. LIMITATIONS

Only one network was retrained for UTSW, and architectures and training configurations were chosen in CV including all sites, so only

the weights were site-naive. At unseen sites, features were z-scored with each site's own label-free statistics; with training-site statistics, features plus age still scored 0.948 (UCSF) and 0.910 (UTSW). The development cohort is 85% grade 4, and site is confounded with prevalence (2% mutant at UPENN, 21% at UTSW), so pooled AUROCs partly reflect grade. The reserved set is case-control enriched (PPV not transferable), and its ensemble–features comparison is post hoc. These patients informed no checkpoint, ensemble choice, combiner or threshold, but entered the CV of two earlier runs comparing training settings. ds007045 tests specificity only. IDH-wildtype status is documented for its original 337 patients, not the 32 added later; excluding these left hybrid specificity at 95.4%. The features are automated surrogates that depend on segmentation (in pooled CV, features plus age scored 0.919–0.944 with automated masks, 574 patients, and 0.894 with manually corrected masks, 288), and the T2/FLAIR ratio is not the visual mismatch sign. Inpainting leaves mass effect. The models target pre-operative MRI. On the earliest post-treatment scan per patient in MU-Glioma-Post [17] (189 patients, 28 mutant), the frozen ensemble did not transfer (AUROC 0.627) and the hybrid (0.870) scored below age alone (0.934).

6. CONCLUSION

On the same patients, age and automated radiological features outperformed a 3D DL ensemble once patients or sites were held out. The ensemble still added T1c information beyond age. The hybrid had the highest observed AUROC in every held-out setting, significantly above features plus age only at UTSW, and was more specific externally than the ensemble or features plus age. DL IDH models should be benchmarked against age and radiological features.

7. COMPLIANCE WITH ETHICAL STANDARDS

This retrospective study used open-access human subject data from UCSF-PDGM, UTSW, UPENN-GBM and MU-Glioma-Post (The Cancer Imaging Archive) and OpenNeuro (ds007045); their licences confirm that ethical approval was not required.

8. ACKNOWLEDGMENTS

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